

Adelowo Adebajji Oluwatimileyin

Applied Mathematician · Scientific Machine Learning

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RESEARCH PROFILE

Applied mathematician with an M.Sc. in Mathematical Engineering, industry ML experience, and independent research in numerical PDEs, scientific machine learning, and inverse problems. Develops verified numerical solvers and controlled comparisons of classical and learned models, with emphasis on convergence, computational efficiency, uncertainty, and generalisation.

RESEARCH INTERESTS

Numerical analysis and PDE-based modelling; scientific machine learning and reduced-order models; inverse problems and uncertainty quantification; computational imaging and fluid dynamics.

EDUCATION

M.Sc. Mathematical Engineering | University of L'Aquila, Italy 2018 – 2021

Thesis: *Bounds on Mixing of Passive Scalars Advected by Incompressible Energy- and Enstrophy-Constrained Flows and Anomalous Dissipation.*

Relevant study: machine learning, scientific computing, optimisation, numerical analysis, finite-element methods, and PDEs.

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria 2013 – 2018

Thesis: *Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.*

RESEARCH EXPERIENCE

Independent Research Projects | *Applied Mathematics & Scientific Machine Learning* Jan 2026 – Present

- Develop and verify spectral and finite-element PDE solvers for incompressible flow, passive-scalar mixing, and phase-field modelling of boiling heat transfer.
- Investigate POD–Galerkin reduced-order models, DEIM hyper-reduction, Fourier Neural Operators, and physics-informed neural networks; evaluate accuracy, runtime, and generalisation to unseen conditions.
- Study inverse problems and computational imaging through Darcy permeability estimation, photoacoustic reconstruction, and variational image restoration.
- Design reproducible numerical experiments using analytical and manufactured solutions, mesh-refinement studies, gradient checks, and held-out, multi-seed evaluation.

RESEARCH METHODS & TOOLS

Python, MATLAB, C++, NumPy, SciPy, PyTorch, FEniCSx/PETSc; spectral and finite-element methods; POD–Galerkin, DEIM, FNO and PINNs; adjoint gradients, optimisation and Bayesian inversion. Verification through manufactured solutions, convergence studies, gradient checks, and multi-seed held-out evaluation.

SELECTED RESEARCH PROJECTS

Navier–Stokes Simulation and Reduced-Order Modeling

- Implemented a Fourier pseudo-spectral flow solver, tensor POD–Galerkin reduction, and DEIM hyper-reduction; verified fourth-order temporal and spectral spatial convergence.
- Rank-16 tensor ROM achieved $38.5\times$ online speed-up at $< 0.1\%$ mean state error on its training trajectory.

Scientific ML: Reduced-Order Models vs. Neural Operators

- Compared POD–Galerkin, DEIM, MLP, and Fourier Neural Operator surrogates for viscous Burgers' equation.
- On 40 in-distribution cases, POD reached 0.152% relative L^2 error versus 0.90% for FNO (five seeds); designed an unseen-family test to characterise model generalisation.

Photoacoustic Reconstruction

- Built a synthetic sparse-view imaging pipeline comparing calibrated time reversal, U-Net refinement, and Tikhonov inversion.
- Across 400 held-out images and five networks, U-Net refinement improved PSNR by 4.9 dB (16 sensors) and 3.2 dB (64 sensors) over time reversal; evaluated reconstruction under controlled sound-speed variations.

Darcy Permeability Inversion

- Implemented adjoint-based Tikhonov inversion and pCN Bayesian sampling; verified the forward solver with manufactured solutions and the gradient with a Taylor-remainder test.
- Conducted sensor, noise, and prior-sensitivity studies with posterior-predictive checks and MCMC mixing diagnostics; used the MAP estimate as the primary reconstruction.

Optimal Mixing of Passive Scalars

- Extended M.Sc. thesis computations with a pseudo-spectral Python solver, reproducing the original tables and adding 48 verification and regression tests.
- A resolution study to $N = 512$ identified moving fit windows as the main source of apparent exponent dependence on resolution; fixed-window fits showed approximately second-order convergence.

INDUSTRY EXPERIENCE

Data Scientist | *Intuos Srl*, L'Aquila, Italy

Dec 2024 – Present

- Develop machine-learning classifiers for aircraft flight-phase identification from recorded operational telemetry.

Machine Learning Engineer | *N-and Italia Srl*, Italy

Apr 2023 – Nov 2024

- Developed predictive models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.

Operational Data Analyst | *eResult Srl*, Cesena, Italy

Jan 2023 – Mar 2023

- Optimised SQL queries and automated recurring business-intelligence reporting workflows.

Digital Engineering Intern | *Bridgestone EMIA*, Rome, Italy

Apr 2022 – Oct 2022

- Developed computational-geometry algorithms and improved data-structure operations for tire cross-section simulation and finite-element meshing workflows.

Machine Learning Intern | *Fellowship.AI*, Remote

May 2022 – Aug 2022

- Implemented an InsetGAN architecture for whole-body image generation and deployed model inference through FastAPI for integration with an iOS application.

SELECTED AWARDS

Professor S.S. Okoya Prize for Excellence in Applied Mathematics (2018); University of L'Aquila International Scholarship (2018–2021).